



School of Computer Science, Engineering and Applications (SCSEA)

B. Tech TY (CCSA)

Subject: Blockchain Systems

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Title of Practical : Set up a private blockchain network using tools like Hyperledger Fabric or MultiChain. Execute basic transactions and smart contracts.

STEP: 1 Prerequisites.

1. Install Git on your system.

```
ubuntu@ip-172-31-91-21: ~  
ubuntu@ip-172-31-91-21:~$  
ubuntu@ip-172-31-91-21:~$ sudo apt-get install git  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
git is already the newest version (1:2.43.0-1ubuntu7.2).  
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.  
ubuntu@ip-172-31-91-21:~$
```

2. Install cURL to fetch files from the internet.

```
ubuntu@ip-172-31-91-21: ~  
ubuntu@ip-172-31-91-21:~$ sudo apt-get install curl  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
curl is already the newest version (8.5.0-2ubuntu10.6).  
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.  
ubuntu@ip-172-31-91-21:~$ |
```

3. Install Docker and Docker Compose to run containers.

```
ubuntu@ip-172-31-91-21: ~/fabric-samples/test-network
ubuntu@ip-172-31-91-21:~$
ubuntu@ip-172-31-91-21:~$
ubuntu@ip-172-31-91-21:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libltdl7 libs1lrp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libltdl7 libs1lrp0 pigz slirp4netns
0 upgraded, 10 newly installed, 0 to remove and 59 not upgraded.
Need to get 121 MB of archives.
After this operation, 442 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libs1lrp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.27-1 [30.5 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.22.0-1-ubuntu.24.04-noble [35.3 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:28.0.4-1-ubuntu.24.04-noble [15.8 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:28.0.4-1-ubuntu.24.04-noble [19.1 MB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:28.0.4-1-ubuntu.24.04-noble [6092 kB]
Get:10 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 2.34.0-1-ubuntu.24.04-noble [14.1 MB]
Fetched 121 MB in 2s (76.6 MB/s)
Selecting previously unselected package pigz.
(Reading database ... 70560 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package containerd.io.
Preparing to unpack .../1-containerd.io_1.7.27-1_amd64.deb ...
Unpacking containerd.io (1.7.27-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../2-docker-buildx-plugin_0.22.0-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.22.0-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../3-docker-ce-cli_5:28.0.4-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-cli (5:28.0.4-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../4-docker-ce_5:28.0.4-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce (5:28.0.4-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5:28.0.4-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:28.0.4-1-ubuntu.24.04-noble) ...
```

STEP: 2 Install Fabric and Fabric Samples.

\$ curl -

sSLO <https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/installfabric.sh>

&& chmod +x install-fabric.sh

```
ubuntu@ip-172-31-91-21:~$
ubuntu@ip-172-31-91-21:~$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
ubuntu@ip-172-31-91-21:~$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

====> Changing directory to fabric-samples
fabric-samples v2.5.12 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

====> Downloading version 2.5.12 platform specific fabric binaries
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.12/hyperledger-fabric-linux-amd64-2.5.12.tar.gz
====> Will unpack to: /home/ubuntu/fabric-samples
% Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 118M  100 118M    0     0  39.7M      0  0:00:02  0:00:02 --:--:-- 41.9M
====> Done.
====> Downloading version 1.5.15 platform specific fabric-ca-client binary
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v1.5.15/hyperledger-fabric-ca-linux-amd64-1.5.15.tar.gz
====> Will unpack to: /home/ubuntu/fabric-samples
% Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 30.0M  100 30.0M    0     0  32.0M      0  0:00:00  0:00:00 --:--:-- 53.8M
====> Done.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccsen baseos
====> Pulling fabric images
====> ghcr.io/hyperledger/fabric-peer:2.5.12
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.48/images/create?fromImage=ghcr.io%2Fhyperledger%2Ffabric-peer&tag=2.5.12": dial unix /var/run/docker.sock: connect: permission denied
```

\$./install-fabric.sh docker samples binary

```
ubuntu@ip-172-31-91-21: ~  
ubuntu@ip-172-31-91-21:~$ ./install-fabric.sh docker samples binary  
Clone hyperledger/fabric-samples repo  
====> Changing directory to fabric-samples  
fabric-samples v2.5.12 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.  
Pull Hyperledger Fabric binaries  
====> Downloading version 2.5.12 platform specific fabric binaries  
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.12/hyperledger-fabric-linux-amd64-2.5.12.tar.gz  
====> Will unpack to: /home/ubuntu/fabric-samples  
% Total % Received % Xferd Average Speed Time Time Time Current  
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
100 118M 100 118M 0 0 37.6M 0 0:00:03 0:00:03 --:--:-- 40.9M  
====> Done.  
====> Downloading version 1.5.15 platform specific fabric-ca-client binary  
====> Downloading: https://github.com/hyperledger/fabric-ca/releases/download/v1.5.15/hyperledger-fabric-ca-linux-amd64-1.5.15.tar.gz  
====> Will unpack to: /home/ubuntu/fabric-samples  
% Total % Received % Xferd Average Speed Time Time Time Current  
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
100 30.0M 100 30.0M 0 0 31.7M 0 0:00:01 0:00:01 --:--:-- 46.3M  
====> Done.  
Pull Hyperledger Fabric docker images  
FABRIC_IMAGES: peer orderer ccenv baseos  
====> Pulling fabric Images  
====> ghcr.io/hyperledger/fabric-peer:2.5.12  
2.5.12: Pulling from hyperledger/fabric-peer  
Digest: sha256:f5f3fb3b061a067df0d13ed109f109cc9cbf02207936438435fcf431a11440  
Status: Image is up to date for ghcr.io/hyperledger/fabric-peer:2.5.12  
====> ghcr.io/hyperledger/fabric-peer:2.5.12  
====> ghcr.io/hyperledger/fabric-orderer:2.5.12  
2.5.12: Pulling from hyperledger/fabric-orderer  
9cb31e2e37ea: Already exists  
904f7cf7cc46: Pull complete  
5a2ed0dfe063: Pull complete  
eeedc9027925: Pull complete  
92d83fce7ae2: Pull complete  
ea198b0fa631: Pull complete  
Digest: sha256:5a91a378ed13de92d1566781dede60baaca42265076f96b5921ad03b25184c6a  
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-orderer:2.5.12  
====> ghcr.io/hyperledger/fabric-orderer:2.5.12  
====> ghcr.io/hyperledger/fabric-ccenv:2.5.12  
2.5.12: Pulling from hyperledger/fabric-ccenv
```

STEP: 3 Run the Fabric Test Network

1. Clone Fabric Samples git repo

\$ git clone <https://github.com/hyperledger/fabric-samples.git>

```
ubuntu@ip-172-31-91-21: ~  
ubuntu@ip-172-31-91-21:~$  
ubuntu@ip-172-31-91-21:~$  
git clone https://github.com/hyperledger/fabric-samples.git  
Cloning into 'fabric-samples'...  
remote: Enumerating objects: 14778, done.  
remote: Counting objects: 100% (259/259), done.  
remote: Compressing objects: 100% (171/171), done.  
remote: Total 14778 (delta 142), reused 88 (delta 88), pack-reused 14519 (from 3)  
Receiving objects: 100% (14778/14778), 23.73 MiB | 24.25 MiB/s, done.  
Resolving deltas: 100% (8091/8091), done.  
ubuntu@ip-172-31-91-21:~$ |
```

2. Go to fabric-samples/test-network

\$ cd fabric-samples/test-network

```
ubuntu@ip-172-31-91-21:~$  
ubuntu@ip-172-31-91-21:~$ cd fabric-samples/test-network  
ubuntu@ip-172-31-91-21:~/fabric-samples/test-network$
```

3. Start the test network

\$./network.sh up

```

ubuntu@ip-172-31-91-21: ~/fabric-samples/test-network
ubuntu@ip-172-31-91-21:~$ cd fabric-samples/test-network
ubuntu@ip-172-31-91-21:~/fabric-samples/test-network$ ./network.sh up
Using docker and docker compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
Unable to find image 'hyperledger/fabric-peer:latest' locally
latest: Pulling from hyperledger/fabric-peer
9cb31e2e37ea: Already exists
433344b70fd2: Pull complete
469a13730448: Pull complete
7a23dc74719b: Pull complete
b8ef412fe6fa: Pull complete
92cf085e65f6: Pull complete
Digest: sha256:995b34aeddbbdc7631460c229d48b5a52b4d388aac73c48df433b4daace9d
Status: Downloaded newer image for hyperledger/fabric-peer:latest
LOCAL_VERSION=v2.5.12
DOCKER_IMAGE_VERSION=v3.1.0
Local fabric binaries and docker images are out of sync. This may cause problems.
/home/ubuntu/fabric-samples/test-network/./bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=../organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=../organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=../organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
WARN[0000] /home/ubuntu/fabric-samples/test-network/compose/compose-test-net.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] /home/ubuntu/fabric-samples/test-network/compose/docker/docker-compose-test-net.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 2/7
  ✓ orderer.example.com Pulled                                1.2s
  ✓ 9cb31e2e37ea Already exists                                0.0s
  ✓ d56d2d9b8d29 Pull complete                                0.1s
  ✓ 3e0332681f56 Pull complete                                0.9s
  ✓ fe41ed8662e4 Pull complete                                1.0s
  ✓ e49cdccbd07e Pull complete                                1.0s
  ✓ c378ba422ddc Pull complete                                1.0s
[+] Running 2/7
  ✓ Network fabric_test Created                                0.1s
  ✓ Volume "compose_peer0.org2.example.com" Created           0.0s
  ✓ Volume "compose_orderer.example.com" Created              0.0s
  ✓ Volume "compose_peer0.org1.example.com" Created            0.0s

```

STEP: 4 The Components of the Test Network

\$ docker ps -a

```

ubuntu@ip-172-31-91-21:~/fabric-samples/test-network$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
46cf46119a3d   hyperledger/fabric-peer:latest      "peer node start"       27 seconds ago Up 26 seconds 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
peer0.org1.example.com
ad139e8b9809   hyperledger/fabric-orderer:latest   "orderer"               27 seconds ago Up 26 seconds 0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
orderer.example.com
46935232b965   hyperledger/fabric-peer:latest      "peer node start"       27 seconds ago Up 26 seconds 0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
peer0.org2.example.com

```

Conclusion: The experiment successfully set up a local Hyperledger Fabric test network using Docker. It provided a working environment to explore core components like peers, orderers, and the CLI, enabling hands-on learning of blockchain operations within Hyperledger Fabric.